Homework 1

ASEN 5335 Aerospace Environment

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Problem 1: Radiation from Corona

(i)

Given T of the corona region to be $T = 2MK$. We can use the equation,

$$\lambda_{max}T = \alpha$$

Where $\alpha = \text{constant}(2.98 * 10^-3K)$.

Rearrange to solve for wavelength.

$$\lambda_{max} = \frac{T}{\alpha}$$

The corona region's temperature can be substituted and solved.

$$\boxed{\lambda_{max} = 1.449 * 10^-9\,\text{m} = 14.49 * 10^-10\,\text{m}}$$

This would also be in the same range for X-rays, $\approx 10^-10\,\text{m}$.

(ii)

For the region photosphere region at a cooler $T = 5,778\,\text{K}$, the process is the same.

$$\boxed{\lambda_{max} = 5.01 * 10^-7\,\text{m} = 50.15 * 10^-8\,\text{m}}$$

This would be in the same range for UV $\approx 10^-8\,\text{m}$.

Problem 2: Thermal Escape

From Wikipedia we can obtain an equation for most probable thermal velocity of a particle.

$$v_{\text{th}} = \sqrt{\frac{2k_BT}{m}}$$

$$k_B = 1.38 \times 10^{-23}\,\text{J/K}$$

$$m = m_{\text{proton}} = 1.673 \times 10^{-27}\,\text{kg}$$

$$T = 2\,\text{MK} = 2 \times 10^6\,\text{K}$$

$$\boxed{\therefore v_{th} = 181,644\,\text{m/s}}$$

We then calculate the escape velocity using the equation,

$$v_{escape} = \sqrt{\frac{2GM}{r}}$$
$$G = 6.674 * 10^{-11}\,\mathrm{Nm^2/kg^2}$$
$$M = 1.989 * 10^{30}\,\mathrm{kg}$$
$$r = 10 * 10^6 * 10^3\,\mathrm{m}$$

Where G is the unversial gravitational constant, M is the mass of the sun, and r is the radius of the of the outer corona. We then find,

$$\boxed{v_{escape} = 163,000\,\mathrm{m/s}}$$

$$\because v_{th} < v_{escape}$$
$$\boxed{\text{It is possible for the coronal protons to escape the sun's gravity}}$$

Problem 3: Solar Wind

(i.)

Given quiescent solar wind density $n = 5\,\mathrm{ions/cm^3}$, and $V = 400\,\mathrm{km/s}$ at Earth, we can use the mass flow equation

$$\dot{m} = \rho * V * A$$

To find A, we calculat the surface area of a sphere given the radius of Earth,

$$A = 4\pi r^2$$
$$r_{Earth} = 149.6 * 10^9\mathrm{m}$$
$$\therefore A = 1.87 * 10^{12}\mathrm{m}$$

Convert the ion density to density in terms of $\mathrm{kg/m^3}$, given the mass of a proton being $m_{proton} = 1.67 * 10^{-27}\,\mathrm{kg/ion}$, we calculate,

$$\dot{m} = 5\,\mathrm{ions/cm^3} * 1.67 * 10^{-27}\mathrm{kg/ion} * \frac{1\mathrm{cm^3}}{1 * 10^{-6}\mathrm{m^3}} * 400 * 10^3\mathrm{m}$$
$$\boxed{\dot{m} = 939,332,865.8\,\mathrm{kg/s}}$$

(ii.)

Finding the flux at the orbit of Jupiter, we are able to use the inverse square relation. We find flux at Earth,

$$\phi_{Earth} = rho * V$$
$$\phi_{Earth} = \frac{5\mathrm{ions}}{1 * 10^{-6}\mathrm{m^3}} * 400 * 10^3\mathrm{m/s}$$
$$\phi_{Earth} = 2 * 10^{12}\mathrm{ions/m^2/s}$$

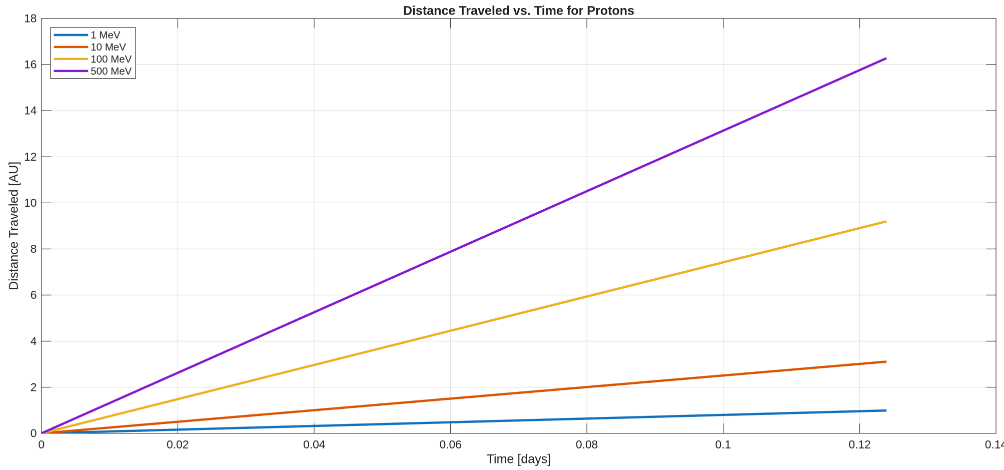

Figure 1: AU travled vs time

With inverse square law if $r_{Earth} = 1\text{AU}$ and $r_{Jupiter} = 5.2\text{AU}$

$$\phi_{Jupiter} = \phi_{Earth} * \frac{1}{5.2}^{2}$$
$$\phi_{Jupiter} = 7.396 * 10^{10}\,\text{ions/m}^2\text{/s}$$
$$\boxed{\phi_{Jupiter} = 7.396 * 10^{14}\,\text{ions/cm}^2\text{/s}}$$

Problem 4: Relativistic Protons

Plotting distance travled in AU w.r.t for multiple protons of varying MeV.

To solve for how long it takes in hours to for each of these energies to each Earth, we are able to use,

$$KE = \left(\frac{1}{\sqrt{1-\frac{v^2}{c^2}}} - 1\right) m_0c^2$$

where we can convert MeV to Joules with,

$$MeV = 1.602 * 10^{-13}\,\text{J}$$

This can be rearranged to,

$$V = \sqrt{1 - \frac{1}{(\frac{KE}{m_0c^2} + 1)^2}}$$

We are able to use this equation to calculate all the velocities given the MeVs. Assuming this velocity is constant through the process, we can simply find some distance d, which is the distance between the lower corona of the Sun and the Earth,

$$d = 1\text{AU} - 2R_S$$
$$1\,\text{AU} = 1.496 * 10^{11}\text{m}$$
$$1\,R_s = 6.957 * 10^{8}\text{m}$$
$$\therefore d = 1.4821 * 10^{11}\,\text{m}$$

Which time t can be calculated with simply $t = d/v$.

	1 MeV	**10 MeV**	**100 MeV**	**500 MeV**
Velocity [m/s]	1.384e+07	4.345e+07	1.285e+08	2.274e+08
Time [s]	1.071e+04	3.411e+03	1.154e+03	6.518e+02
Time [hr]	2.975	0.947	0.320	0.181

Table 1: Properties of Lower Corona Proton at varying MeV

Problem 5: Heliopause

In order to calculate the heliopause we first seek the pressure as a result of the velocity and ion density at Earth. The inverser square law can then be applied to solve for the heliopause distance.

$$P_{Earth} = n_{Earth} m v^2$$
$$n_{Earth} = 5\,\text{ions/cm}^3 = 0.005\,\text{ions/m}^3$$
$$m = 1.673 * 10^{-27}\,\text{kg/ion}$$
$$v = 400 * 10^{3}\,\text{m/s}$$
$$\therefore P_{Earth} = 1.3384 * 10^{-21}$$

(Problem 6: Spacecraft Temperature)

(i.)

Given aluminum with $\alpha = 0.15$ and $\epsilon = 0.05$, to find equalibrium temperature at Earth, we use $r = 149.6 * 10^9$m and then use the equation for thermal equalibrium,

$$T = (\frac{\alpha}{\epsilon})^{\frac{1}{4}} (\frac{S A_n}{\sigma A_{tot}})^{\frac{1}{4}}$$
$$S = 1361\,\text{W/m}^2$$
$$\sigma = 5.67 * 10^{-8}\,\text{W/m}^2/\text{K}^4$$
$$A_n = \pi(149.6 * 10^9)^2\,\text{m}^2$$
$$A_{tot} = 4\pi(149.6 * 10^9)^2\,\text{m}^2$$
$$\boxed{\therefore T = 366.3\,\text{K}}$$

Where S is solar flux, σ is the Stefan-Boltzmann constant, A_n is the cross sectional area normal to the Sun, and A_{tot} is the surface area at Earth's orbit.

(ii.)

For Jupiter, we modify the r used to be the r at Jupiter, however use the inverse square relation to find the new solar flux S.

$$S_{Jupiter} = S_{Earth} * (\frac{1\,\text{AU}}{5.2\,\text{AU}})^2$$
$$S_{Jupiter} = 50.3\,\text{W/m}^2$$

The process remains the same substituting $r = 778 * 10^9\,\text{m}$

$$\boxed{\therefore T = 160.6\,\text{K}}$$

(iii.)

When repeating the process for GSFC White Paint NS-74 with $\alpha = 0.17$ and $\epsilon = 0.92$, we get,

$$\boxed{T = 182.5\,\text{K}}$$

I think this paint was designed with the explicit purpose of being able to have a lower equalibrium temperature as a result of radiation from the Sun for satellite missions. High temperatures can be detrimental to equipment onboard and the strategy to paint the spacecraft with paint that improves its emmissivity by a considerable amount would be very useful in the space environment by lowering the equalibrium temperature.

Problem 7: Solar Outputs

(a.)

Based on the Figure provided, the minimum radiative energy flux of an M-Class flare would be $10^{-5}\,\text{W/m}^2$.

(b.)

Assuming the peak energy of the flare being $\phi_{1AU} = 5 * 10^{-4}\,\text{W/m}^2$, we can find the corresponding energy flux at the emmision source at 1.5 solarradii to be a result from the inverse square law.

$$\frac{\phi_{1AU}}{\phi_{1.5solarradii}} = (\frac{1.5solarradii}{1AU})^2$$
$$1.5_{solarradii} = 1.04 * 10^9$$
$$1AU = 1.496 * 10^11$$
$$\text{We rearrange for } \phi_{1.5solarradii}$$
$$\phi_{1.5solarradii} = \phi_{1AU}(\frac{1.496 * 10^11\,\mathrm{m}}{1.04 * 10^9\,\mathrm{m}})^2$$
$$\boxed{\therefore \phi_{1.5solarradii} = 0.0719\,\mathrm{W/m^2}}$$

(c.)

According to this chart, the pattern of a sharp rise and slow decay over time seems to occur only twice, once on the 7th and once again on the 13th. Meaning that there was likely 2 CMEs launched into the Heliosphere in this time period.

(d.)

When comparing the relation between X-ray flares and the flux of solar protons, notice that when there is a X flare event, there is an increase in solar proton flux.

(e.)

When comparing the relation between cosmic rays and proton flux and radiative energy flux, we can see that when the solar proton flux decreases over time as a result of an X-Flare, the cosmic rays measured would be negative of (% of Baseline), meanwhile near the start where it is at more of a steady conditon, there is a positive and more steady (% of Baseline)

(Problem 8: Parker Solar Probe)

My favorite instrument aboard the Parker Solar Probe would be the FIELDS insturment. It's purpose is to measure magnetic fields of the solar corona during the PSP mission to study the Sun. FIELDS has two fluxgate magnetometers, one serach coil magnetometer, and five voltage sensors. It is accompanied by four whip antennas. This can then measure Electric Field, Magnetif Field, EM waves, and RF emmisions using this combination of antennas and magnetometers, being able to sample at 2 MHz. It's SWAP would depend on how one classifies the size of the instrument. The accompanying antennas are approximately 2 meters, and the magnetometer boom is 3.5 meters, though the size of the magnetometers are about the size of a fist. As a whole it would weigh roughly 23 kg, and roughly 22 watts. The instrument contributes to the PSP's parent investigation of characterizing the dynamic region close to the Sun by being the one to take the field measurements with its antennas and set of magnetometers.